A Systems Engineering Approach to the Performance Enhancement of a Thermal Vacuum Chamber

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Part I. Needs Analysis

A number of deficiencies as well as technological opportunities relating to the Thermal VAcuum Chamber (T-VAC) have been identified to warrant the need for an improved T-VAC system within the CubeSat facility. The system was first created by Justin Foley about 15 years ago when TRW, now part of Northrop Grumman, donated the chamber to CubeSat and has not been updated since its first iteration.

One such flaw regards the old Data AcQuisition system (DAQ) which would often report sporadic temperature readings. This is extremely detrimental to the ability to provide appropriate testing conditions as well as customer satisfaction. Another major deficiency is that testing data could be lost if the computer recording system would spontaneously shut down. As the system often requires a large amount of power to run, the circuit breakers would activate at times thus shutting down the recording system as well. This would cause a test to have to be restarted to recreate the data. In one extreme case, a test that should have taken three days to complete took nearly three weeks. A third identified flaw is the unstable thermal control system. The system at times even ignores a command entirely which makes it difficult to accurately control the testing environment. The resulting inefficiency is that testing can take longer to complete in order to achieve the appropriate condition and also a decrease in customer satisfaction since the ability to maintain a stable temperature had been compromised.

There are also some technological advancements that opens opportunities to advance the T-VAC system significantly. Primarily, the improvement of computer technology allows for a higher processing rate of temperature as well as temperature control. This would assist in creating a more stable and responsive environment for the T-VAC system. Also, improvements in DAQ hardware and particularly analog to digital converters, allows for more accurate temperature measurements. Lastly, improvements in software for reading DAQ hardware has made it much easier to display and record data for the user.

It is important to determine the feasibility of any system before entertaining the idea of creating it. With some preliminary research, it was found that many industries use similar technologies for their T-VAC systems. While some improvements to the system, such as replacing the entire chamber, are not feasible due to the cost, other, smaller improvements, such as replacing the computer and DAQ are feasible and have been done in the past with other systems similar to the one at the CubeSat facility.

It is also important to understand and visualize the primary objectives and the resulting functions derived from those objectives. An objectives tree is a useful tool for these purposes:

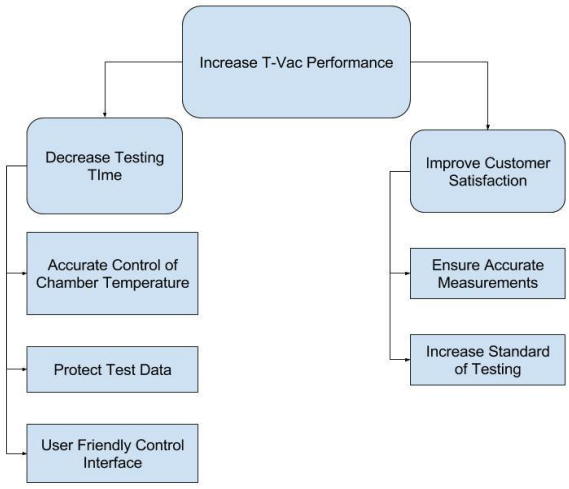


Figure 1: Objectives Tree

Overall, the goal of the T-VAC is simply to perform valid thermal vacuum test within the parameters of the MIL-STD-1540 and GSFC-STD-7000 standard. While the previous T-VAC system did perform this basic objective, there are two major operational objectives for improving the T-VAC chamber: customer satisfaction and decreasing testing time.

Customer satisfaction is essentially at the core of any product or service provided and T-VAC operations are no exception. A number of incidents could have resulted in a decrease in customer satisfaction such as inaccurate measurements and the standard of the testing environment. Customers rely on consistent measurement taking in order to validate their equipment. Also, there are times when the temperature measurement were obviously wrong or started transmitting due to a DAQ hardware issue.

Decreasing the testing time is another area that would greatly improve the performance of the T-VAC chamber. The testing facility is primarily run by students who also have school work to complete and decreasing the testing time would allow students to also focus on their school activities. Moreover, decreasing the testing time makes the testing process more efficient by decreasing overhead costs. First, protecting the testing data in case of power failure will help decrease the testing time. Due to the high power requirements of the T-VAC pump processes, there have been instances where the circuit breakers have activated and shut down power to the entire system including the computer

recording the data. Unfortunately, this caused some data to be lost as it is not automatically saved on the system. Another functionality that would decrease the testing time is the accurate control of the testing chamber temperature. By doing so, the user will be able to get to and from hot and cold soaks faster. Third, a user friendly control will help expedite the testing process by having the status of the chamber clear and concise terms.

Part II: Concept Exploration

Knowing the main functions of the improvements, one can begin formulating some performance requirements based upon the operational objectives. These operational objectives have been chosen specifically to address each of the issues mentioned in the needs analysis phase.

Improve Customer Satisfaction	Decrease Testing Time
Ensure Accurate measurements: • The system shall record temperature values within 0.1º C of true value	Accurate Control of Chamber Temperature: • The system shall display a temperature time derivative • The system shall employ a pulse with modulation scheme
Increase Standard of Testing Environment: • The system shall maintain a maximum pressure of 1E-4 torr during testing	Protect Test Data: • The system shall automatically record and save data on regular intervals • The system shall be able to function for a minimum of five minutes with electrical outlet power failure.
	User Friendly Control Interface: • The system shall pass internal user reviews • The system shall be able to be further modified for future updates

Table 1: System Performance Requirements

One can also begin defining system subsystems and components described in the diagram below:

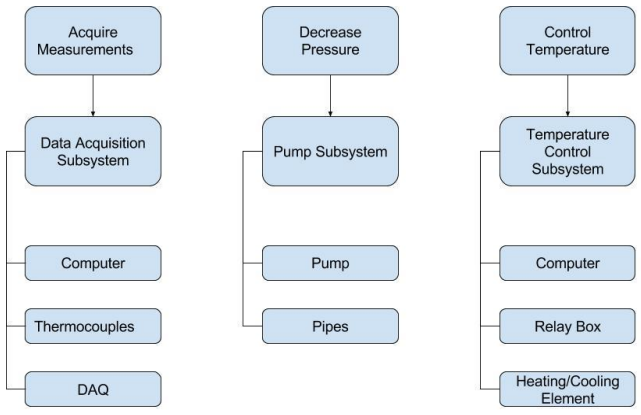


Figure 1: Subsystem and Component Diagram

It is interesting to note that, in Figure 1, the computer component is displayed in two separate subsystems. This is an early indicator that this component is essential to binding subsystems together. By defining some of the component level systems that will be included into the system, one can start developing some preliminary system concepts. Being that the overarching goal is a system upgrade as opposed to an entirely new system, one can focus on the component level to either upgrade or use the component as is. However, the question of software choice is also essential to the development of different candidate concepts. There are a few options to the type of software to implement to the system. Some options include: buying computer software, buying hardware with reintegrated software, or developing the software internally.

Part III: Concept Definition

With system performance requirements and candidate system concepts, one can begin developing a functional and physical architecture and analyze the candidate systems to derive system functional definitions and defined system concepts. First, a functional analysis must be done to determine what components perform which functions.

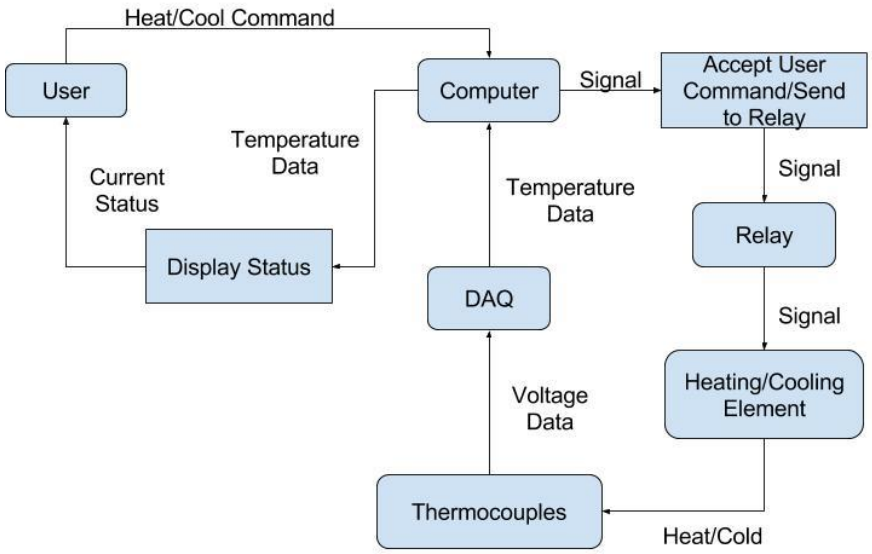


Figure 2: Functional Block Diagram

As Figure 2 shows, the main process of the system is a feedback loop between the computer and the user to control the temperature. However, in the process of applying some of these upgrades, it was found that some of the components were taking on dual roles in the functional block diagram. For example, when the DAQ was upgraded to a completely new piece of hardware, it was at the time unknown that the DAQ was also serving as the signal

board for the relays. This caused some problems since a new method of sending commands to the relay board was needed. Ultimately, a new method of giving signals to the relay board had to be devised.

Next, a physical definition helps determine which components to retain and which to upgrade. These trades incorporated multiple criterion such as cost, feasibility, complexity, schedule cost (time to implement), and lifetime/robustness. While the ability to further upgrade these systems is a requirement, it was important to the customer to have a long lifetime to avoid the need for another upgrade in the near future. It is also assumed that all hardware can eventually be replaced.

Component	Decision	Justification
Computer	Upgrade	Easily Upgradable for low cost
DAQ	Upgrade	Needed to meet system requirements
Pump	Retain	Costly to upgrade, not necessary to meet requirements
Pipe	Upgrade	Ability to upgrade without buying new components, keeps cost low
Relay Box	Upgrade	Needed to meet system requirements
Heating/Cooling Element	Retain	Extremely expensive to replace, integrated into entire chamber

Table 2: Overview of Trade Study for Component Upgrades

The goal of the trades shown in Table 2 was primarily to get the largest impact from improvements to the system while keeping costs reasonably low. It is important to note that the lack of information known about the current system had some effect on the level of improvements able to be made. For example, is it not entirely known how the heating/cooling element is integrated into the chamber and was therefore deemed to be too much of a risk to the system integrity to try to try to de-integrate that component.

There was also a trade done for what software implementation technique should be used for the system. While either buying or purchasing software pre-integrated into hardware would be easiest, it was decided that producing the hardware internally would best fulfill the requirement of ensuring the software in upgradable in the future. Many of the decisions were made using the stage gate process in which an idea was reviewed and was either passed, rejected, or sent back for further analysis by the members of the CubeSat facility.

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A Systems Engineering Approach to the Performance Enhancement of a Thermal Vacuum Chamber

Engineering and Post Development

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IME 510-01

Thermal testing and qualification is integral to any spaceflight mission and is also a requirement before hardware is allowed to be flown into the space environment. This paper explores the upgrades performed on a Thermal VACuum (T-VAC) chamber facility from the perspective of the systems engineer focusing on the engineering and post development stages of the process. Such a global approach is fundamental to the successful production of a complex system that includes both hardware and software components because it ensures that the changes to one particular subsystem does not cause undesirable effects within other subsystems. Also, a systems engineer's global perspective allows one to decide the most efficient improvements to be made to the system to ensure that the requirements, voice, and wants of the customer are satisfied. Another paper entitled "Systems Engineering Approach to the Performance Enhancement of a Thermal Vacuum Chamber" discusses the concept development of the T-VAC system and the associated upgrades that needs to be implemented to gain the most benefit while staying within budget. Some of the requirements to improve customer satisfaction and increase test efficiency include employing a pulse with modulation scheme, passing internal user reviews, calculating the time derivative of temperature, and the ability for software to be modified in the future. The engineering required to achieve these requirements is discussed in this paper as well as the implications of some engineering decisions made to achieve a higher standard of testing for customers.

Part I: Advanced Development

Some new technologies were produced in order to meet requirements including a solid state relay box, a LabVIEW user interface, and a LabVIEW to Arduino communications protocol. As one of the functional requirements states: the T-VAC system needs to be able to reliably and accurately control the application of heat or cold to the chamber. While mechanical relays implemented in the old system do provide control of the heater and cooler, it was found that they failed often upon extended, repeated use and required a relatively long duty cycle on the order of seconds to maintain operational. Therefore, Solid State Relays (SSRs) are employed instead to better meet requirements. SSRs have considerably expanded capabilities in that they have a much longer lifetime and lower risk of failure since they do not carry any mechanical components. They also have the added advantage that they can employ a Pulse With Modulation (PWM) scheme that tricks the heaters to hold an intermediate heating level instead of being completely on or off. This essentially supplies an analog signal to the heaters instead of a binary on or off signal that in turn gives much finer control of the heat input to the system. As SSRs have not been employed to control heaters from a LabVIEW user interface before, the technology to produce this functionality had to be created. Arduinos are well documented, open source, and relatively inexpensive micro controllers that have an immense amount of existing documentation and user base. It is therefore natural to use this microcontroller to provide the signal for the SSRs that are being used. Preliminary research was done as to the type of protocol that SSRs use to function and relays were

bought that accept a 3-32 V DC input since this voltage can be implemented with the Arduino's 5 V output.

With the new technologies being developed to meet system requirements for the T-VAC upgrades, one needs to address the risks that are associated with the new technologies being implemented. One tool to describe these risks is to create a risk matrix as in Table 1 to determine the most alarming risks. The most alarming risk shown in this table is a software error as that would have the extensive severity of stopping thermal data collection and as the software is relatively new and untested, the possibility of such an occurrence is relatively high. As the software is used and tested over time, the possibility of an error occurs decreases as well as the users trust in the software. To mitigate the possibility of error during the early stages of testing however, the program is given simulated data and allowed to supply commands to the Arduino LED testbed (discussed in Integration and Evaluation) before the software is implemented in the system.

Table 1: Risk Identification

Severity	Extensive	Rapid Unexpected Pressurization		Software error		
	Major		DAQ error	Power Failure		
	Medium			Thermocouple Failure		
	Minor					Run out of LN2
	No Impact					
		Highly Unlikely	Unlikely	Possible	Likely	Very Likely
	Probability					

In order to place priority onto the mitigation of these risks so that the most predominant risks are addressed first, a risk priority table is created and examined as in Table 2. One finds that running out of Liquid Nitrogen (LN2) is actually the highest risk for the T-VAC chamber as running the chamber at high vacuum without it can not only damage the article being tested, but the testing hardware as well. Also, it is nominally expected for more than one tank of LN2 to be used so the likelihood of LN2 running out is very high. With regards with new technology being implemented however, the incident with the highest risk priority number is a Data AcQuisition (DAQ) error. This is because the ability to read the temperature of the thermocouples and collect data is the primary directive of the testing facility. If the DAQ hardware were to suffer some error, this functionality would be lost and the facility would have to come to a complete halt until a solution was found.

For some of these issues, mitigation techniques already exist in the non-upgraded system. For example, before the system runs out of LN2, it is simply replaced with a full tank. The DAQ error occurs when there is miscommunication between the new DAQ hardware and the computer or the DAQ connection to the thermocouples is lost. Such an error

Table 2: Risk Priority

Incident	Severity	Probability	Ability to detect (Low Number = High ability to Detect)	Risk Priority Number
Rapid Unexpected Pressurization	10	1	2	20
DAQ Error	9	2	4	72
Software Failure	9	2	2	36
Power Failure	7	6	1	42
Run Out of LN2	3	9	3	81
Thermocouple failure	5	4	2	40

can be fixed with a list of troubleshooting processes listed including rebooting the device, checking connections, and restarting the software. In case a single thermocouple fails, redundant thermocouples are placed on critical test points so that the area can still be tested in the event that one fails, another one can still collect the necessary data. Additionally, the likelihood of a Rapid Unexpected Pressurization was determined to be so low that the only mitigation techniques that were deemed necessary was to ensure that the chamber is properly sealed off before beginning the vacuum process. Unfortunately, facility power failures are becoming more common as the power requirements of our equipment becomes higher. To mitigate this, alternate plug configurations exist in case one plug configuration does not work.

Software Systems

In order to meet requirements, an application software was created that would use an Arduino to accept and interpret commands from LabVIEW and implement the appropriate corresponding functions to the relay box. First, a communication protocol was defined between the LabVIEW computer software and the Arduino. That protocol is then interpreted into different functions for how much power the relay box should be providing the heater or what duty cycle the cooler should be implementing. The data sent to the Arduino is a string of char bytes along with a predefined delimiter and terminator. The Arduino is programmed to look for the delimiters and terminator and also understands what each part of the command stands for. The communication protocol is summarized in the table below.

Table 3: Communication Protocol

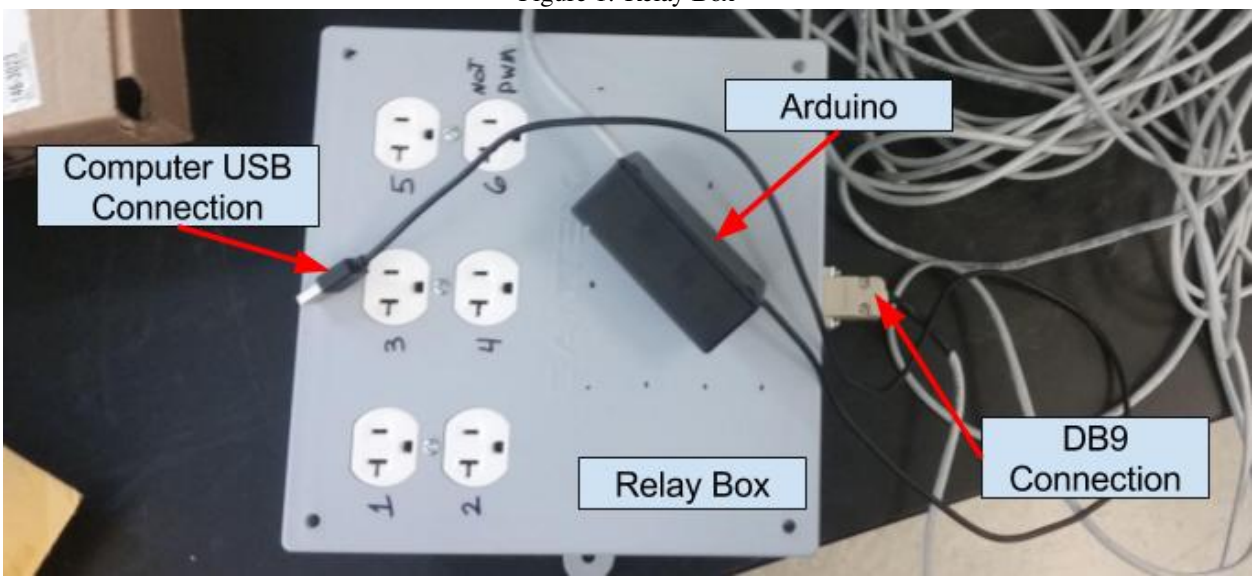
Definition	Plug definition or "PERIOD"	On or Off	Duty Cycle	Terminator: Defined as '%' character
Description	Indicates which heater/cooler to control or indicates a change of period	Indicates whether the heater or cooler is on or off	Defines the duty cycle for the heater or cooler	
Example	'H1' 'C' 'H2'	'0' '1'	'0' '50' '65.7'	

The LabVIEW software also provides other functions such as displaying a recording data a graphical display proved to be the most useful when displaying temperature data since it proved the user a quick way to gauge the difference between temperatures between different parts of the testing environment. A simple numerical display is useful for the pressure data since there is no comparison for the pressure, just to ensure that it does not raise above a certain pressure predetermined depending on the test being done.

Part II: Engineering Design

Prototypes and evaluations plans were created for the components and subsystems that were being upgraded. Such components include the Arduino, Computer, and Relay Box. The effectiveness of these subsystems and components are critical to achieving the PWM requirements mentioned earlier.

Figure 1: Relay Box



Arduino

As mentioned in "Systems Engineering Approach to the Performance Enhancement of a Thermal Vacuum Chamber", there was an unexpected consequence of upgrading the DAQ being that the GPIO functionality does not exist with the new DAQ that was purchased to meet functional requirements. Therefore, an Arduino Leonardo is implemented to provide General Purpose Input Output (GPIO) functionality to the relay component. On a hardware level, it was decided that the best way to ensure a reliable communication link is to have a physical wire run from the Arduino to the relay box. This was done by using CAT-8 Ethernet wires with DB9 ports since both components are already well tested for reliability by industry.

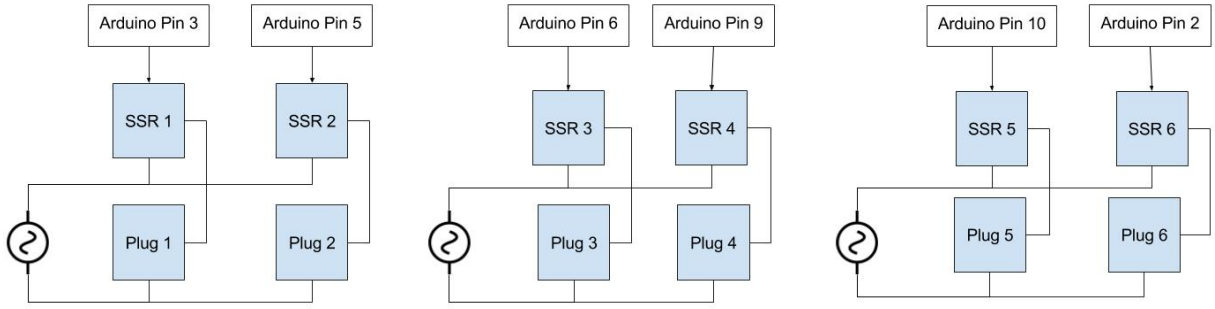
Relay Box

The objective of the relay box is to house the SSRs that output power through a traditional plug that fluctuates according to the PWM value given by the user though the computer and Arduino. There are many considerations keep in mind such as space, heat, and user indication during the design and engineering of the relay box. Because of customer demand, all the outlets, wires, relays, and lights have to fit inside the relay box. This forced a design that used the lease amount of wires and the most condensation of components. The relays were packed as tightly as possible and wiring was daisy-chained as much as possible to conserve space. However, to meet the power requirements of the heaters being used in the testing facility, only two plugs could be daisy chained at a time. Thermal concerns due to the large amount of power being flown through the relays required the need for heat sinks on every relay and thermal paste was applied to encourage heat transfer from the relays to their heat sinks. Next, a large aluminum plate was placed below the box to allow the heat to escape from the heat sink as well as to provide a stable base for the box itself. Lastly, LED indicators were placed on the relay box itself so that a user would have immediate indication as to the operational status of the relay box. This includes one LED indicator for each of the three AC plugs as well as an LED indicator for each of the outlets on the relay box to indicate which outlets are being used.

Computer

According to requirements, a user friendly environment needs to be created for the user to control T-VAC. The LabVIEW software environment was chosen over other software development environments because of its ease of creating graphical user interfaces and existing legacy software that has been used for the facility. Another requirement, as mentioned earlier, is the ability to calculate the time derivative of the temperature to give the user a better idea of the

Figure 2: Relay Box Connection Schematic



state of the chamber. In order to fulfill this requirement the computer subsystem requires a hardware upgrade to do the calculations required in a timely fashion. Fortunately, a computer that meet the memory and processing power demands was found within the Cal Poly surplus system and CubeSat was able to successfully request a transfer of that computer for CubeSat use. It features 4 GHz of processing power as well as 16 GB of RAM which exceeds the minimum processing capabilities needed to run the LabVIEW environment.

Part III: Integration and Evaluation

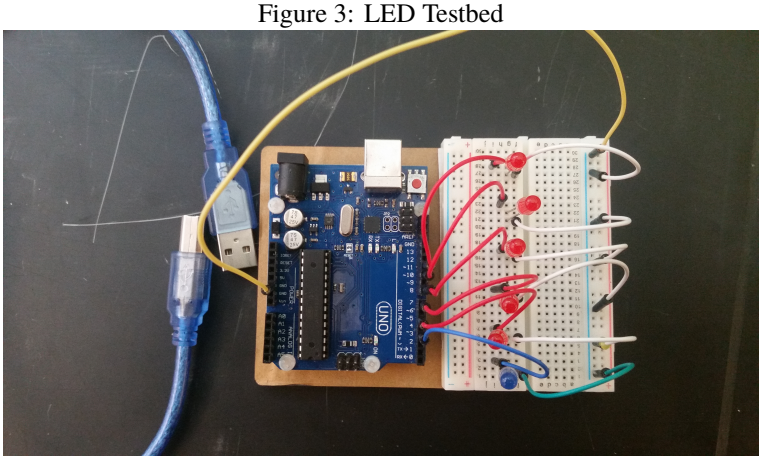
There were a series of verification and validation tests performed to ensure that the updates were meeting the objectives and standards set forth during the Concept Development phase. Preforming these tests proved to be extremely helpful to the successful implementation of the upgrades into the system because several bugs were found that were fixed in testing that would have been disastrous if said malfunctions occurred during actual testing.

Communication Protocol

A verification test had to be performed for the new communication protocol being implemented between the LabVIEW software and the Arduino. An initial verification test consisted of the Arduino repeating any command it received. Following the initial verification test, the command was broken into its component commands that were repeated separately and once the communication protocol was successfully verified to be working, the system can be validated. Validation tests were performed on the Relay Subsystem consisting of the LabVIEW Software, Arduino, and the relay box itself before implementing the subsystem into the full T-VAC system. An initial validation test was performed using a LED that simulated the powering on and off of the relays. LEDs provide a suitable simulation since they are compatible with a PWM scheme and only require a 5 V source so there is little risk of harm to the test environment

or tester. Once the Arduino software passes the LED testbed, the software system can be implemented on the actual relay box.

Figure 3: LED Testbed



Relay Box

Testing the relays verifies the high power functionality of the relay box alongside the PWM scheme implemented by the software in conjunction with the SSRs. First, a single relay was tested but failed to turn on when given a signal from the Arduino. As it was the first time that such a test was being performed, it was at first assumed that the wiring must be incorrect. However, a multimeter sweep ensuring that the physical connections proved that they were correctly wired. To verify that the relay would respond to any input, a power source was used to activate the relay. It was found that the relay required 5.5 V to activate as opposed to the advertised 3 V. The unit was returned and the same type of relays were purchased from another manufacturer and were verified to be in working order. After, more relays were added to the Arduino and controlled simultaneously. Finally, 120 V AC power was provided through the relay and tested with an outlet plug and a lamp. The goal of the test was to control the lamp plunged into the outlet that was in turn connected to the relay. The test proved to be successful and integration of the relay box could begin.

The integration of the relay box into the larger system after verification and validation tests were simply a "plug-and-play" design. Since the relay box does not carry any programming logic or software, the only integration concerns for the box are with regards to the internal hardware. One concern that was brought up during integration of the box was the fact that foreign objects such as dust and bugs might have a way to enter and disrupt the electronics. To mitigate this, the cover was placed on with the gasket such that there were no openings for foreign objects to enter. Also, tape was placed over some holes at the bottom on the relay box that were created to mount the SSRs but were not used. This covered the remaining foreign object entry points.

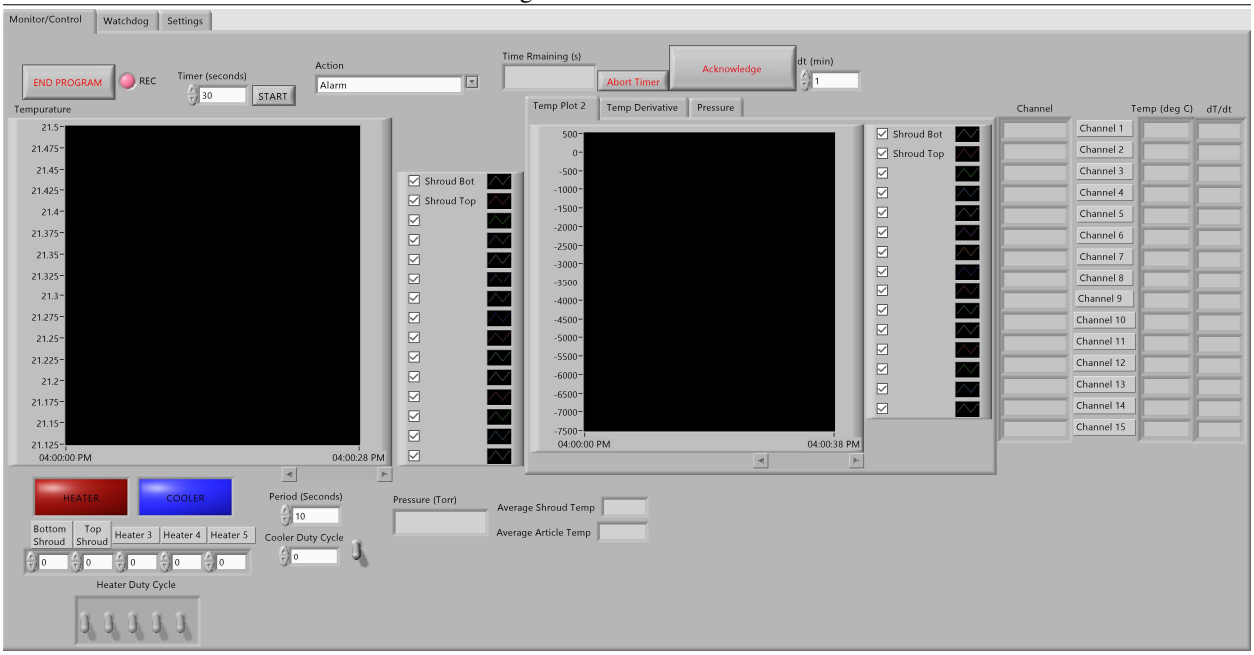
Data Acquisition

The integration of the DAQ and the software into the new system had to be simultaneous. This is because the old hardware was not compatible with the new software being implemented in the new system. This made it difficult to track any malfunction down to a single sub-system or component. When first trying to implement the new software alongside the new hardware, there were many times where the LabVIEW software would not recognize the hardware. At first, it was assumed that the cable between the hardware and the computer was faulty since it was an in house product. However replacing the cable did not solve the issue. Eventually is was found that the LabVIEW software was simply not configured correctly to accept the new hardware and once the software was appropriately configured, both systems functioned within normal parameters.

User Interface

Another, more abstract, validation that occurred was the LabVIEW computer software's requirement to be "user-friendly." Many meetings were held to observe the layout and functionality of the user interface and experience. Many utilities such as check boxes, tabs, and "at-a-glance" indicators proved to enhance the user experience and make T-VAC testing go smoother overall. Many times throughout the production of the software, changes to requirements and functionality requests occurred. In order to implement these requests, two rules were employed to improve the quality of the software while maintaining functionality. First, any upgrades to the software had to maintain any previous functionality, unless the upgrade specifically called for the end of a certain function. This practice allowed for the software development to maintain fidelity in case the software needed to be used while functionalities were still being changed. The second general guideline used to ensure the stability of the software during development is that upgrades had priority numbers. This way, users can decide on which updates they felt were most important and the software would be changed to accommodate the most useful changes first. As mentioned earlier, before any software was committed to the system, it was tested with simulated thermocouple data and allowed to give commands to the Arduino LED testbed. This verification test proved to be fruitful as there were some LabVIEW software bugs found between the communication protocol of the computer software and the Arduino.

Figure 4: User Interface



Part IV: Production

The product being supplied by the T-VAC facility and the upgrades implemented is the ability to qualify and test flight vehicles for launch. Therefore, the production phase focuses on the ability to produce such tests for space vehicles. The upgrades completed were designed to facilitate the increased efficiency of testing as well as increase customer satisfaction by providing more accurate and continuous data. The requirements in "A Systems Engineering Approach to the Performance Enhancement of a Thermal Vacuum Chamber" were specifically chosen to do just that. Therefore, it is reasonable to assume that if requirements were met, the performance of the production, namely qualification testing, should increase. The validation and verification tests done to the subsystems ensured that requirements were being met and that the subsystems being developed are designed to meet requirements. Fortunately, the facility has been used after the writing of this paper and it was found that the testing efficiency and ease of use did, in fact, increase. This is a major benefit to the CubeSat Program as the reduction in testing time in turn decreases overhead cost such as power and water consumption as well as employee work hours.

The upgrades help reduce power costs through the implementation of the SSRs. The SSRs allow a consistently lower wattage to be used by the heater instead of being constantly being turned on and off at full power. Water is also being conserved since it is used to cool down the pumps. Naturally, the less time spend preforming a test, the less time the pumps needs to be used and therefore the less water used to cool down the pumps. Lastly, the employees used for T-VAC testing are students who also have other academic responsibilities. The ability for a test to complete in a

timely manner is important to the students' ability to perform in school.

Part V: Operations and Support

There are significant operations and support considerations for the upgrades as well as the original parts of the system. For example, one operation concern is the fact that the system pump uses water as a coolant and has to remain on after testing has completed. If unchecked, the water can remain on for a long time and can cause environmental and financial concerns. Also, in order to maintain the performance of the system, new members of CubeSat are being taught the backend code along with troubleshooting directions in the event malfunctions occurs.

A manual regarding the operation of the Hardware and User Interface. These manuals allow a person to operate the T-VAC chamber without previous knowledge of the system as well as a reference guide for operators who already are familiar with the system. The manual also includes troubleshooting procedures for the software, DAQ, and the pump system. A changes to the hardware occur, this manual is updated, reviewed by staff, then submitted for future use and reference. Making the manual a living document as opposed to a static one allows for an easier flow of information from senior members of CubeSat who have created the changes to the system to junior members.

Documentation on the code developed within LabVIEW and the Arduino. These references are geared for CubeSat employees who wish to make further upgrades to the software system. Such documentation is critically important to the longevity of the software used. If there is no documentation explaining how and why the code operates in the way it does, when software bugs come in the future, CubeSat may decide to rebuild the program again from scratch instead of fixing or improving upon the current software. This would likely result in a loss of functionality until the rebuilt software adds the functionality.

The Systems Engineering process has been integral to the success of the upgrades preformed on T-VAC Chamber. The ability to create a conceptual template for an engineered design, preceded to develop the design, then follow through with production and product support has improved the efficiency and customer satisfaction of the T-VAC Chamber immensely. However, there are still more improvements to be made to further improve the T-VAC chamber. One such improvement is to automate the heating and cooling of the test article through the computer instead of having a human user input the control parameters. Another improvement entails providing the computer with the pressure data so that it can be recorded along with the temperature data in a more accurate manner. Altogether, the upgrades made in this iteration of the product support process was been very successful and if the support cycle for the facility continues in a manner that is beneficial to maintaining the continuing improvements, one can expect the

facility to continue to run for many years to come.

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